

2017 量子 725

一.

(1) Stark 效应: 原子或分子在外电场作用下能级和光谱发生分裂的现象

Zeeman 效应: 原子在外磁场中发光谱线发生分裂且偏振的现象

$$\begin{aligned} (2) \quad [\hat{L}_x, \hat{L}_y] &= [\hat{y}p_z - \hat{z}p_y, \hat{z}p_x - \hat{x}p_z] = [\hat{y}p_z, \hat{z}p_x] - [\hat{y}p_z, \hat{x}p_z] - [\hat{z}p_y, \hat{z}p_x] + [\hat{z}p_y, \hat{x}p_z] \\ &= [\hat{y}p_z, \hat{z}p_x] + [\hat{z}p_y, \hat{x}p_z] = \hat{y}[\hat{p}_z, \hat{z}]\hat{p}_x + \hat{x}[\hat{z}, \hat{p}_z]\hat{p}_y \\ &= -i\hbar\hat{y}\hat{p}_x + i\hbar\hat{x}\hat{p}_y = i\hbar(\hat{x}\hat{p}_y - \hat{y}\hat{p}_x) = i\hbar\hat{L}_z \end{aligned}$$

同理, 可得 $[\hat{L}_y, \hat{L}_z] = i\hbar\epsilon_{ypr}\hat{L}_r$

(3) $E_n \sim n^2$: 一维无限深方势阱

$E_n \sim n$: 一维线性谐振子

$E_n \sim \frac{1}{n}$: 类氢原子

$$(4) \quad i\hbar \frac{\partial}{\partial t} U(t, 0) = \hat{H} U(t, 0) \Rightarrow U(t, 0) = e^{-\frac{i}{\hbar} \hat{H} t}$$

$$\langle \hat{A} \rangle(t) = U^\dagger(t, 0) \hat{A} U(t, 0) = e^{\frac{i}{\hbar} \hat{H} t} \hat{A} e^{-\frac{i}{\hbar} \hat{H} t}$$

$$\begin{aligned} \frac{d\langle \hat{A} \rangle}{dt} &= \frac{i}{\hbar} \hat{H} e^{\frac{i}{\hbar} \hat{H} t} \hat{A} e^{-\frac{i}{\hbar} \hat{H} t} + e^{\frac{i}{\hbar} \hat{H} t} \frac{\partial \hat{A}}{\partial t} e^{-\frac{i}{\hbar} \hat{H} t} + e^{\frac{i}{\hbar} \hat{H} t} \hat{A} \left(-\frac{i}{\hbar} \hat{H} \right) e^{-\frac{i}{\hbar} \hat{H} t} \\ &= \frac{i}{\hbar} e^{\frac{i}{\hbar} \hat{H} t} (\hat{H} \hat{A} - \hat{A} \hat{H}) e^{-\frac{i}{\hbar} \hat{H} t} + e^{\frac{i}{\hbar} \hat{H} t} \frac{\partial \hat{A}}{\partial t} e^{-\frac{i}{\hbar} \hat{H} t} = \frac{i}{\hbar} e^{\frac{i}{\hbar} \hat{H} t} [\hat{A}, \hat{H}] e^{-\frac{i}{\hbar} \hat{H} t} + e^{\frac{i}{\hbar} \hat{H} t} \frac{\partial \hat{A}}{\partial t} e^{-\frac{i}{\hbar} \hat{H} t} \end{aligned}$$

$$\Rightarrow \frac{d\langle \hat{A} \rangle}{dt} = \frac{i}{\hbar} \langle [\hat{A}, \hat{H}] \rangle + \left\langle \frac{\partial \hat{A}}{\partial t} \right\rangle$$

(5) 能量增加, $E_n \sim n^s$ ($1 < s < 2$) 且当 $|x|$ 越大, $s \rightarrow 2$.

$$(6) \quad \vec{A} = (-By, 0, 0)$$

$$\hat{H} = \frac{1}{2m} \left(\hat{p} - \frac{q\vec{A}}{c} \right)^2 = \frac{1}{2m} \left[\left(\hat{p}_x + \frac{qBy}{c} \right)^2 + \hat{p}_y^2 \right] = \frac{\hat{p}_x^2}{2m} + \frac{1}{2m} \left(\frac{q^2 B^2}{c^2} \right) \left(y + \frac{c\hat{p}_y}{qB} \right)^2$$

(7) 处于激发态的发光原子在外场的作用下, 向低能态或激发态跃迁时, 辐射光子的现象

偶极跃迁的选择定则: $\Delta l = \pm 1$, $\Delta m = 0, \pm 1$

二.

$$(1) \quad -\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} + V\psi = E\psi \quad \Rightarrow \quad \begin{cases} \psi'' + k^2 \psi = 0 & |x| \geq b \\ \psi'' + p^2 \psi = 0 & |x| \leq b \end{cases} \Rightarrow \psi = \begin{cases} e^{ikx} + re^{-ikx} & x \leq -b \\ Ae^{ipx} + Be^{-ipx} & |x| < b \\ te^{ikx} & x \geq b \end{cases}$$

$$\Rightarrow \begin{cases} \psi(b) = \psi(b) \\ \psi'(b) = \psi'(b) \\ \psi(-b) = \psi(-b) \\ \psi'(-b) = \psi'(-b) \end{cases} \Rightarrow \begin{cases} Ae^{ipb} + Be^{-ipb} = te^{ikb} \\ ipAe^{ipb} - ipBe^{-ipb} = ikte^{ikb} \\ e^{-ikb} + re^{ikb} = Ae^{ipb} + Be^{-ipb} \\ ik e^{-ikb} - ik re^{ikb} = ipAe^{ipb} - ipBe^{-ipb} \end{cases}$$

$$\text{移动坐标轴, 令 } -b \rightarrow 0, \quad b \rightarrow a \quad V(x) = \begin{cases} 0 & x < 0, x > 2a \\ -V_0 & 0 \leq x \leq 2a \end{cases}$$

$$\psi = \begin{cases} e^{ikx} + re^{-ikx} & x < 0 \\ Ae^{ipx} + Be^{-ipx} & 0 \leq x \leq 2a \\ te^{ikx} & x > 2a \end{cases} \quad \text{解得} \quad \begin{cases} r = \frac{(k^2 - p^2) \sin(pa)}{(k^2 + p^2) \sin(pa) + i2kp \cos(pa)} \\ t = \frac{2kp e^{-ik2a}}{(k^2 + p^2) \sin(pa) + i2kp \cos(pa)} \end{cases}$$

$$\Rightarrow R = |r|^2 = \frac{(k^2 - p^2)^2 \sin^2(pa)}{(k^2 - p^2)^2 \sin^2(pa) + 4k^2 p^2} \quad T = |t|^2 = \frac{4k^2 p^2}{(k^2 - p^2)^2 \sin^2(pa) + 4k^2 p^2}$$

$$(2) \quad 2b = n \cdot \frac{\lambda}{2} = n \cdot \frac{1}{2} \frac{2\pi\hbar}{p}$$

$$\therefore p = \frac{n\pi\hbar}{b} = \sqrt{\frac{2m(E - V_0)}{\hbar^2}} \Rightarrow E = \frac{n^2 \pi^2 \hbar^2}{8mb^2} - V_0 \quad (n = 1, 2, \dots)$$

三.

设零级近似波函数为 $\psi^{(0)} = C_1 |1\rangle + C_2 |2\rangle$

$$\text{有 } \begin{pmatrix} H'_{11} & H'_{12} \\ H'_{21} & H'_{22} \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = E^{(1)} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} \quad \text{即} \quad \begin{pmatrix} H'_{11} - E^{(1)} & H'_{12} \\ H'_{21} & H'_{22} - E^{(1)} \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = 0 \quad \Rightarrow \quad \begin{vmatrix} H'_{11} - E^{(1)} & H'_{12} \\ H'_{21} & H'_{22} - E^{(1)} \end{vmatrix} = 0$$

$$\text{其中 } H'_{11} = \langle 1 | H' | 1 \rangle = \omega_1 \langle 1 | e \rangle + H.C. \quad H'_{12} = \langle 1 | H' | 2 \rangle \quad H'_{21} = \langle 2 | H' | 1 \rangle \quad H'_{22} = \omega_2 \langle 2 | e \rangle + H.C.$$

$$\text{代入, 解得 } E^{(1)} \text{ 和 } E^{(2)} \quad \therefore E_1 = E^{(0)} + E^{(1)} \quad E_2 = E^{(0)} + E^{(2)}$$

$$\text{代回解得 } C_1, C_2 \quad \psi^{(1)} = \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}$$

四.

$$(1) \psi_{100} = \sqrt{\frac{2+12}{\pi a_0^3}} e^{-\frac{3r}{a_0}}, \quad a_0 = \frac{\hbar^2}{m e^2}$$

$$(2) P = |\langle \psi_{10} | \psi_{100} \rangle|^2 = \left| \int_0^{\infty} \sqrt{\frac{2+12}{\pi a_0^3}} \cdot \sqrt{\frac{2}{\pi a_0^3}} e^{-\frac{3r}{a_0}} r \cdot 4\pi r^2 dr \right|^2$$

五.

(1) L_z 的可能值为 $\hbar, 0$, 相应概率为 $|a|^2$ 和 $|b|^2$

$$(2) \hat{L}_x = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \quad \hat{L}_x \psi = E \psi \quad \psi = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

$$\Rightarrow \hat{L}_x = \hbar \text{ 时 } \psi_1 = \frac{1}{2} \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix} \quad \hat{L}_x = 0 \text{ 时 } \psi_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \quad \hat{L}_x = -\hbar \text{ 时 } \psi_3 = \frac{1}{2} \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix}$$

$$\therefore [\hat{H}, \hat{L}_z] \quad \hat{H} \text{ 的本征值 } E_1 = \hbar \mu_B \quad E_2 = 0, \quad E_3 = -\hbar \mu_B$$

$$\text{令 } \mu_B = \omega, \quad \psi(t) = a \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} e^{-i\omega t} + b \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a e^{-i\omega t} \\ b \\ 0 \end{pmatrix}$$

$$\therefore \hat{L}_x = \hbar \text{ 时 } P_1 = |\langle \psi_1 | \psi(t) \rangle|^2 = \frac{1}{4} |a e^{-i\omega t} + \sqrt{2} b|^2$$

$$\hat{L}_x = 0 \text{ 时 } P_2 = |\langle \psi_2 | \psi(t) \rangle|^2 = \frac{1}{2} |a|^2$$

$$\hat{L}_x = -\hbar \text{ 时 } P_3 = |\langle \psi_3 | \psi(t) \rangle|^2 = \frac{1}{4} |a e^{-i\omega t} - \sqrt{2} b|^2$$

六.

$$(1) \text{基态 } \psi = \psi_0(x_1) \psi_0(x_2) |0, 0\rangle \quad E = \frac{\pi^2 \hbar^2}{4 m b^2}$$

$$\text{第一激发态 } \psi = \begin{cases} \frac{1}{2} [\psi_0(x_1) \psi_1(x_2) + \psi_0(x_2) \psi_1(x_1)] & |0, 1\rangle \\ \frac{1}{2} [\psi_0(x_1) \psi_1(x_2) - \psi_0(x_2) \psi_1(x_1)] & \begin{cases} |1, 1\rangle \\ |1, 0\rangle \\ |1, -1\rangle \end{cases} \end{cases}$$

$$E = \frac{5\pi^2 \hbar^2}{8 m b^2}$$

$$(2) \vec{H}' = -\vec{M} \cdot \vec{B} = \frac{e \hbar m_s \hbar}{m c}$$

$$\text{当 } m_s = -1/2 \text{ 时 } \frac{e \hbar \hbar}{m c} = \frac{3\pi^2 \hbar^2}{8 m b^2}, \quad B_z = \frac{3\pi^2 \hbar}{8 e b^2}$$